

Atom processor have been catering to these mobile platforms. While it has certainly been profitable for Intel, the existing technology hasn't exactly been a star performer, as was previously expected and has failed to make an impression among the consumers. It has also failed to quell the growing importance that ARM devices have been starting to get in the recent years. Haswell is not only Intel's first serious attempt at acknowledging the importance of the growing market share of ultra mobile portable devices, it also attempts to bring architectural consistency across all product lines upto server platforms.

The most important factor that has prompted Intel to focus on developing solutions for mobile platforms is users' increasing tendency to move away from desktops and laptops towards devices such as tablets, all-in-ones and detachable form factors. It is also a fact that Intel no longer enjoys the same position that it did in the desktop market a few years ago. There have even been talks about desktop sales declining in favour of these new devices. It is also a growing reality that ARM has been one of the universally accepted platforms for smartphones and mobile devices on leading platforms like Android and iOS. In light of all these factors, it does make sense for Intel to revisit its strategy from the ground up and jump into the market with a solution that pulls the consumers in its direction.

The new series of processors cater to three broad categories i.e. Desktop, Server and Mobile platforms. The desktop series start with the low power and ultra low power S & T series and go up to the K series unlocked processors which offer an adjustable CPU multiplier. These K series parts however do not support Transactional Synchronization Extension (TSX) and Virtualization (VT-d). The mobile processors, on the other hand, have a broad range as well, with its variants including M, U, Y and H series processors. While the M series denote mobile processors with Quad core chips, the H series offer a Quad core chip along with Intel Iris Pro graphics solution. The U and Y series, on the other hand, are low power devices targeted for ultrabooks, tablets and other similar devices.

FULLY INTEGRATED VOLTAGE REGULATOR

The new Haswell processors now have a fully integrated voltage regulator (or FIVR) within the CPU itself, which is a boon for most devices. On one hand, it does take away a lot of freedom

from the motherboard manufacturers, while on the other, this integrated FIVR's performance outruns any other voltage regulator that works externally. Also voltage regulation can be done to a much finer degree across all motherboards. Additional benefits include much faster transition times and a reduction in the number of components on the motherboard. It is obvious that this has been done with mobile devices in mind, as the reduction in motherboard components will also allow for smaller form factors.

NEW SOCKETS

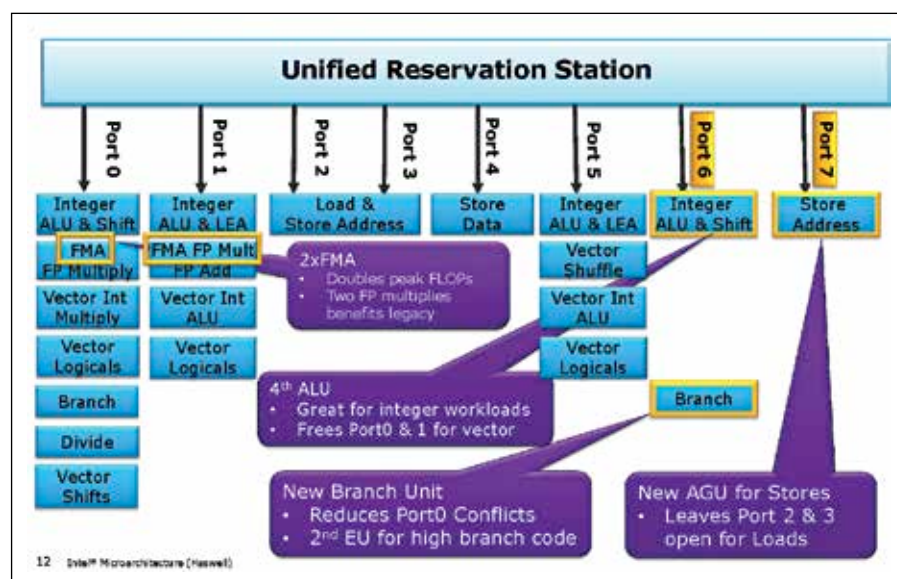
With the advent of Haswell, Intel will be moving to a new 8 series chipset and will be using the new LGA1150 socket for most devices. There will be a BGA packaging option for a limited set of chips labelled as the R series of devices.

POWER CONSUMPTION

Another innovation that these new devices have required is in terms of power consumption. Standby power consumption is one metric that stands apart when compared to previous chips. According to official figures, the standby battery life has more than doubled for the Haswell CPU. Ivy Bridge had two power states i.e. Awake (S0) and Sleep (S3/S4). Now, Haswell adds a whole new category of idle states called SOix. When the CPU is in one of these states, it produces idle characteristics, while at the same time making the OS believe that it is in an active state and can switch from idle to active state almost instantaneously, at a speed much faster than before. This not only gives us an improvement in the battery life, but is also completely

transparent to well written software. There are multiple such Active Idle states which promise lower power consumption depending on the activity of the user, for example SOi1 and SOi3 which are used between periods of activity and inactivity of the user. The shift to these Active Idle states becomes important in light of the new paradigm of always on devices which need to always stay connected and offer longer battery life.

The new platform also offers more features for tackling power management, which includes Platform Activity Alignment and Panel Self Refresh. Platform Activity Alignment allows for the OS to schedule the data transactions between components in such a way that they coalesce together to enable the system to wake up only in short bursts of time instead of it being done as soon as the component is ready with the data. This allows the CPU to sleep for a longer period of time. Panel Self Refresh, on the other hand, attempts to save battery from the display, which needs to be refreshed almost 60 times in a second even if something is not changing on the screen. PSR proposes a solution in which the display panel can come with a DRAM attached to the display panel; this will store the frame buffer until a time arrives when the display needs to be changed. This allows the device to decouple with the display and wake up less often, thus saving even more precious power. There's another improvement, selective PSR, that can allow saving of more power from displays. It allows updating of only a selected portion of the display, as long as the rest of the background remains the same. The



The Haswell Execution Unit Improvements